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The crisis in academic statistics: an exploration

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Abstract

At present, statistics as an *academic* career is not only experiencing an identity crisis, but is also fighting for survival as a vital and vibrant career choice. Good financial benefits, as well as the increasing demand for *applied* statisticians, have opened up very attractive career opportunities for statistics graduates, while at the same time eroding the viability of pursuing statistics as an academic teaching profession. As a result, the number of students wishing to teach statistics at universities is dwindling, and the number of vacancies in statistics departments at higher education institutions has increased.

This article explores some of the reasons for the above, as well as the impact of the crisis on academic statistics. The article considers possible solutions – like opening the discourse on the future of academic statistics, taking students' experiences seriously, the Africanisation of the curriculum, the effective use of technologies, ongoing research, and adequate financial reward.

Keywords: Academic (theoretical) statisticians, non-academic (applied) statisticians, higher education

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Introduction

Both nationally and internationally there is an urgent need for well-trained and qualified statisticians, especially in higher education, as statistical knowledge is becoming increasingly important and in demand. Very large data sets have to be collected properly and analysed using scientifically justified methods. Lamel (2003: 383) states that "Impartial, relevant and reliable statistics are among the cornerstones of democratic societies". Within the specific context of Africa, there are clear expectations that South Africa will take the lead to address the "dramatic declines and diminished levels of competence in its [Africa's] statistical systems" (Lehohla, 2008: 2). Already in 2003, the Minister of Finance of South Africa lamented

> the lack of trust in statistics, poverty of statistical skills, mediocre discourse where results trump methods, and the onerous requirements of playing in a "higher league" of statistical practice, where the International Monetary Fund's special data dissemination standards set the benchmark (Lehohla, 2008: 2).

Higher education plays a crucial role in addressing the need for more and better skilled statisticians (Lindsay, Kettenring & Siegmund, 2004; Smith & Staetsky, 2007). Yet the number of academic statisticians in higher education is fast declining. This article considers some of the possible reasons for this phenomenon, as well as the impact of this crisis on academic statistics. Should the present trend continue, higher education will not be able to provide society with enough well-trained and suitably qualified applied statisticians, because there simply will not be sufficient academic statisticians to train them.

This article aims to

- provide a brief overview of the current crisis in academic statistics
- explore statistics as a profession (academic and applied statisticians)
- draft a profile of a typical academic statistician
- formulate recommendations and suggest further research

Problem statement

Vacancies in the statistics departments at higher education institutions are on the increase, while undergraduate student numbers in a variety of statistics courses are on the rise. The situation no longer does justice to the reasonable requirements of statistics as a discipline, and has indeed become a crisis in higher education. Who will teach these students?

While the number of postgraduate students is declining, those who do qualify envisage careers as applied statisticians or mathematicians, or wish to apply their knowledge in a range of related disciplines, for instance the social sciences, economics and en-

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gineering. The young academic statistician of today can move relatively easily from being a "pure" statistician into one of these related fields. Statistics forms part of the infrastructure of a range of different sciences, and universities often exploit the interdisciplinary nature of the subject in curriculum choices. As other options become more attractive (and often more lucrative), there has been a resulting decrease in the number of students who wish to pursue statistics as an *academic* teaching career. The decline in the number of students choosing an academic career in statistics is further exacerbated by mainly two other factors, namely the ongoing identity crisis in statistics, and the increasing role of technology in, and its impact on, statistics.

Literature review

The present crisis in academic statistics and the impact of the crisis on the future of academic statistics are well-theorised in the international domain (see, for example, Barabba, 1990; Kettenring, 1997; Moore, 2001), but less so in the African and specifically South African contexts. In the South African context, the research focus in statistics has been mainly on the teaching of the subject (see, for example, North & Zewotir, 2006; Zewotir & North, 2007), and not on theorising about the content and future of academic statistics. A literature review of international trends in academic statistics does, however, confirm present concerns in the South African higher education context.

Kettenring (2001: 1229), for instance, states that "The need for statistical thinking has never been greater" and that "No group is in a better position to crack open the challenges of the information age than we are". Despite these positives, "statistics seems to be struggling" (Kettenring, 2001: 1229). Moore (2001: 1) summarises the crisis as follows: "Although the discipline of statistics is healthy, its place in the academe is not". Already in 1990 Barabba (1990: 125) stated that, while the number of undergraduate students taking statistics has dramatically increased, the number of postgraduate students has decreased significantly.

A literature review of the reasons for this crisis indicates that it has arisen, on the one hand, from the historical position of statistics as a discipline (see Kettenring, 1997; Moore, 2001) and, on the other hand, as a result of the impact of new technologies on statistics (see, for example, Bickel, 2004; Moore, 1997; Pike, 2002).

If one looks more closely at the situation, it emerges that, firstly, there are several factors *internal* to statistics as a discipline contributing to the present crisis (Green, 2003: 424; Moore, 2001). From the inception of statistics as a separate discipline, its relationship with mathematics has been central to the challenges it faces (Cobb & Moore, 1997; Moore, 2001). Moore (2001: 2) explores the difference between statistics and mathematics, but then states that both disciplines need each other. He describes the two

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disciplines as "natural allies" (2001: 2). Moore (2001: 2–3) proposes that a functional alliance between the two disciplines can benefit both. Opposing such an alliance is Friedman (2001: 9) who states that statisticians must "moderate [their] romance with mathematics," and Straf (2003: 4) who recommends empirical co-operation rather than theoretical, mathematical validation. Another perspective is provided by Moore and Cobb (2000: 615). They investigate tensions and co-operation between statistics and mathematics, not only comparing instruction within the disciplines, but also examining the environments in which the two fields operate. These authors formulate a thesis that the cultural strengths of statistics could greatly assist mathematics, while the organisational strengths of mathematics could provide shelter for statistics, a situation that may prove to be essential for its survival. Fragmentation of the discipline is one of the reasons why many universities dissolved their statistics departments, and statisticians had to find a home in other departments, such as computing, some of the social sciences and the business sciences. To counter the crisis, Kettenring (1997: 1229–1234) proposes an "image reconstruction" of academic statistics. As part of this "image reconstruction" Kettenring proposes, among other things, that statistics own up to the negative image the discipline has in popular discourse, that concerted efforts be made to reach out to women and retired statisticians, that opportunities are created for international cooperation in the teaching of statistics, that the interdisciplinary character of statistics is defined and celebrated, that there is innovative thinking to cope with the increasing size of datasets and, lastly, that the infrastructure in statistics as a discipline is re-imagined (see also Moore, 2001 for different solutions). Gnanadesikan (1990) proposes the optimisation of cross-disciplinary opportunities for statistics, while suggesting that the erosion of the discipline-specific nature of statistics is one of the dangers inherent in the growth of interdisciplinary applications (see, for example, Moore & Olkin, 1984).

The second source of the identity crisis of statistics lies in the fact that statistics is considered to be a technology rather than a separate discipline in its own right. The idea that statistics may best be considered a technology rather than a science is distressing, considering that the study of statistics has become a way of developing a deeper awareness of the world around us. As early as 1978, Healy (1978: 385) posed the question whether statistics is indeed a science. Bickel (2004: 408) states that it should be acknowledged that people in all mathematical sciences are interested in learning from data and, therefore, the often paranoid efforts to distinguish statistics from other fields should stop. The increase in technological applications moves statistical inference back towards its roots, enriching the subject, but data analysis requires a high level of technological proficiency on the part of students. Statistics as a discipline can only survive through acceptance of a different and broader view of its scope (Straf, 2003: 2). Only through change can statistics play what has been termed "a leadership role in confronting the challenges of technological change". With the increase in the number of technologies available in the field of data interpretation, the internal identity crisis experienced by academic statisticians is far from over.

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The identity crisis in statistics as a separate discipline could either drive the professional statistician into further isolation, and into eventually becoming an endangered species, or the situation could be changed into an opportunity to create a high-class career option. Becoming an academic statistician is not, however, an uncomplicated career choice. Higher education should urgently confront the need for more academic statisticians by taking the complexities and uncertainties associated with the current situation into account.

Within this broader context of the identity crisis of statistics and the impact of new technologies, we will now continue to analyse statistics as a profession.

An analysis of statistics as profession

The scope of the statistical profession has expanded as a consequence of the development of a large number of statistical techniques during the past century. As a result, statistics as a discipline has become divided into a wide variety of specialised fields, although the members of the profession are all called s*tatisticians* irrespective of the field in which they operate. The general statement that the need for statisticians is increasing, applies to the need for qualified *applied* statisticians (Lehohla, 2005). However, it applies equally to the need for *academic statisticians*, a need which is indeed increasing because universities are having difficulties in filling vacant teaching posts.

For the sake of clarity, it is perhaps apposite at this juncture to distinguish between academic and non-academic statisticians. This distinction does not, however, imply that statisticians who operate in non-academic professional environments lack an academic background, nor that statisticians in the academic community are ill equipped or unable to apply their knowledge. This distinction rests on the teaching of statistics as a profession as opposed to the range of other possible professions in which statisticians may engage. The teaching of statistics as a *career* is losing prominence, while the application of statistics in other professions is being actively pursued. It is necessary to consider the essential differences between the academic and the non-academic statistician, and find a way in which the gap between the two can be closed, or at least narrowed**.**

The academic statistician

Teaching statistics used to be based on theoretical and practical knowledge of the underlying mathematical principles. The academic statistician still needs to view statistics and mathematics as natural intellectual allies in the debate on the relative roles of thought and automation (Moore, 2000: 129). In the words of Moore and Cobb (2000: 616), "[m]athematics offers statistics intellectual as well as institutional protection". Cheung (1998: 3) contends that the academic statistician knows the value of abstract mathematics, has a keen interest in research, and is sometimes little concerned with messy, real-world problems. The latter allegation is a serious one, implying that

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academic statisticians inhabit ivory towers. If the allegation is true, then academic statisticians who, for instance, leave messy, real-world problems to applied statisticians, should not complain when the oxygen in the ivory tower starts to run low. The stark distinction between theory and application has become outdated and the need for "pure" theoretical knowledge obsolete (Barnett, 2000: 409).

The non-academic statistician

Non-academic statisticians, or applied statisticians, are interested primarily in the ability to use mathematical concepts as tools for achieving fluency in the use of applications. They are less interested in the specialisation of the subject and the abstract foundations of their applications.

The application of statistical knowledge by suitably qualified professionals is very much in demand. One such professional is the *official* statistician, whose task it is to provide information that acts "as a sort of highway on which most of the other statisticians circulate when they perform their work" (Martin-Guzmán & Cervera, 2002: 1). Official statisticians play a very important role in government offices, and globalisation has extended their responsibilities far beyond the boundaries of their home environment.

In the past, official statisticians have provided the basis for the quantitative assessment of the state, society and the economy, and they will continue to do so in the future. The historical use of this form of statistics dates back to ancient times, when populations were counted for governance and taxation purposes. Modern official statisticians face new demands from a variety of users for the multiplicity of uses to which official information is put. In 2003, the President of South Africa asked higher education institutions to become involved in the quest to train more statisticians (Mbeki, 2003). In his request for a strong cadre of statisticians, he was referring to the need for *applied* statisticians, and in particular official statisticians in the state apparatus.

At present the demand for official statisticians exceeds supply. This often results in the appointment of unqualified or under-qualified people, who are liable to interpret data either unethically or incorrectly. Maxims such as "most people use statistics the way a drunk uses a lamp post, more for support than enlightenment" originate as a response to the statistical practices of unqualified or under-qualified persons and officials.

Applied statisticians find employment not only as official statisticians, but also in the economic and social environments, where statistical analysis has become a significant component of strategic planning and development. Statistical literacy has become trendy, especially in the world of business. Small businesses consult independent statisticians, and larger companies have statisticians on their payroll. As a rule, the statistician in the world of applied statistics earns much more than the academic stat-

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istician, and has more opportunities and scope. Although some academic statisticians are also involved in applied statistics, their employment contracts severely restrict their involvement in this area.

Considering the work of academic and non-academic statisticians, it is obvious that *both* are professionals. Both professions require extensive knowledge of the underlying theoretical mathematics and, at the same time, need the technical expertise and soft skills necessary to contribute effectively to society, industry and government. However, in practice statisticians are very seldom sufficiently skilled in all these areas. The practical statistician seems to see the theoretical basis of applications as irrelevant, while the academic statistician is not always interested in market-determined uses for statistics.

This rather discouraging divergence between academic and non-academic statistical streams has existed since the nineteenth century. Various efforts have been made to bridge the differences between the two. At the time of the establishment of the International Statistical Institute (ISI) in 1885, academic statisticians showed little interest in the practical problems of the official statisticians, who, for their part, rejected complex mathematical tools (Cheung, 1998: 3). The aim of the ISI was to provide a discussion forum where academic and official statisticians could meet, exchange knowledge and achieve consensus so as to improve estimates and reduce errors in the work of the latter.

The organisation, Statistical Partnerships among Academe, Industry and Government (SPAIG) (http: //www.svsu.edu/orgs/spaig/), advocates the restructuring and redirection of graduate education in statistics, with special emphasis on traditional doctoral training of academic statisticians. Of particular interest is the theoretical research training of the academic statistician, which focuses too narrowly on technical abilities. The training lacks an emphasis on non-technical soft skills, such as communication and an understanding of the statistician's role in industry and government.

Despite these efforts, it seems that the statistical community has not been able to solve the problem of differences and disparities between academic and non-academic statisticians. This may be one of the reasons for the increasing rarity of the academic statistician as a species. Problems encountered in practice in this age of transparency of financial institutions and globalisation should provide fertile areas for research by academic statisticians, whose area of expertise should be the development of new conceptual frameworks and new innovative analytical methods, such as model-based approaches, to obtain reliable estimates. The application of Bayesian techniques, Kalman filtering and other sophisticated techniques by official statisticians may result in eventual collaboration between academic and official statisticians.

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The non-academic statistician, however, is flourishing, while the responsibility to modify and foster an interest in statistics as an independent discipline seems to rest solely with the academic statistician. Pike (2002) sees the twenty-first century as an age of assessment, of quality assurance and of accountability. In twenty-first century societies, then, the role of statisticians is crucial, as theirs is the task of critically engaging with an abundance of information and data and transforming this information (usually numerical) into usable knowledge. This task fits perfectly with the description of the non-academic statistician, but it also traces a full circle back to the trainer of these non-academics – the academic statistician who must identify and communicate both *why* and *how*.

Based on the literature review and analysis, we will now set out to draw up a profile of the 'typical' academic statistician. Having clarity on the profile will enrich the discourse on academic statistics, and will also assist in defining specific interventions.

Profiling the typical academic statistician

There has been a general decrease in student enrolments for degrees with a major in statistics. The mathematical base of statistics is relatively abstract and theoretical and, because of a general lack of mathematical and numerical literacy among students today, students tend to favour other disciplines. Current curriculum development in statistics as a discipline increasingly emphasises non-mathematical approaches and the effective use of computer software (see, for example, North & Zewotir, 2006; Ridgway, Nicholson & McCusker, 2007). While such approaches may benefit students at an elementary level, they do not deal with the need for an increase in the number of students who take statistics as a major. The decline in student numbers could also be the result of industry dissatisfaction with university courses which are regarded as not practical enough. Patience is not a virtue of the business sector, and business and industry do not always understand that re-curriculation at tertiary level is a lengthy process. Industry prefers instead to introduce a variety of custom-made short courses and to make in-service training available.

Students graduating with a major in statistics can continue with postgraduate studies and, should they be interested in joining the ranks of academic statisticians, the minimum requirement is a Master's degree in statistics. For such a candidate to enter the academic sector at the lowest level as a junior lecturer, and to rise through the ranks to a full professorship, further qualifications are needed. Compounding the challenges faced by academic statisticians, are the exigencies of coping with the current general shortage of teaching staff, the increasing emphasis on research and publication, continuous curriculum renewal, and growing demands made by students. Universities are currently supportive of the ambitions of junior academics, but very few academics are available to occupy the more senior academic positions (and thus attract better remu-

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neration). Consequently, young graduates usually leave the academic sector before or just after the completion of a Master's degree or PhD to take up attractive employment offers from government or industry.

In both South Africa and other countries many statistics departments have merged with either mathematics or commerce departments. Often academic positions are vacant for long periods, either because no applications are received or because applicants do not satisfy the specified criteria. The current staffing in higher positions at many universities consists either of older people or foreigners (for a discussion of the situation in the UK, see Smith & Staetsky, 2007). In the lower ranks, a great deal of movement to non-academic professions has a negative impact on the functionality of a department. Once a department becomes understaffed, the remaining members must constantly move between the various specialities of the discipline while coping with impractical work allocations. Under these circumstances, it is not unusual for those academic statisticians who are able to transfer to the non-academic sector to make use of the opportunities on offer. The problem therefore only escalates further.

Over and above the vocational disadvantages of being an academic statistician, the reality of teaching statistics at a tertiary institution has become more challenging than ever before.

Recommendations and suggestions for further research

Cloning an academic statistician

The current considerable interest in applied statistics is praiseworthy and, although the discipline is healthy, educated decisions about its place in academia are essential, as the discipline is losing ground, except for its growth in basic service courses. Moore (2000) asks us to lower our expectations when he concludes that statistics in practice has moved away from mathematics, that statistics embedded in other degrees may be more realistic than an aspiration to establish it as a major, and that undergraduate programmes need not train professional statisticians. One may use cloning as a humorous explanatory tool. If the cloning of academics statisticians were at all a realistic and possible solution, the following would be a good indication of the required character and competencies of the ideal academic statistician.

Clones would need to be gifted with logical reasoning preceding application, interpretation and estimation – all this clothed in willpower, brainpower, perseverance, enthusiasm and commitment. Pike (2002: 2) adds technical competence and charisma as characteristics of the academic statistician. Clones would also need to have a talent for business, since "selling" the discipline is essential. In South Africa, those responsible for cloning statisticians would have to take cognisance of the following issues:

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- *Gender.* There is a need for more women, and more specifically, women from previously disadvantaged backgrounds. The discipline is still dominated by white males.
- *A passion for teaching.* Academia cannot easily match the remuneration associated with being a non-academic statistician. The reasons for being in the academic world and not in a better-paid position in industry must be researched and celebrated. Even if the remuneration packages that higher education could offer to statisticians were to match those offered by industry, a passion for teaching would be a defining characteristic and requirement.
- *Effective use of technology.* The need is for an innovative and up-to-date approach to, and knowledge of, technology to the point of being aware of speculation about quantum computers and the anticipated breakthroughs that they will make possible. Above all, the art would be to make students aware of the use of different technologies, even while many of them may not yet have access to these diverse forms of technology.
- *Being human.* Straf (2003: 3) suggests that, as fallible human beings, statisticians fail because of personal influences and social attitudes and values. There is therefore a need for a regulatory and training environment that emphasises ethical and responsible conduct.

Cloning as a means of resolving the crisis in academic statistics is not a new idea. In discussing efforts to satisfy the need for more suitably qualified law practitioners, Ome (2003) discusses the "cloning" of a number of law practitioners. The "cloning" was a success, but resulted in unforeseen and often hilarious impracticalities. A case in point was when the original members and their respective clones had the same interests and needs, and asked for the same salaries. The Law School had to file for bankruptcy!

Opening the discourse

One of the possible sources of the present crisis in statistics is the fact that statistics as a discipline is not static – compared with what it was some four decades ago, for instance, it is now almost unrecognisable. Nevertheless, the academic part of the discipline remains enclosed in an ivory tower. Academia has excluded industry, society and students from participating in the discourse about the nature of the discipline and its role in contemporary society. Knowledge production in the twenty-first century has become a contested enterprise where the academic world, industry, the state, as well as society at large, all stake different claims. Instead of an eerily silent and unattainable ivory tower, academia should become a noisy agora, or meeting place, where the different stakeholders can converse and share ideas.

Naïve statistical applications in industry are a reality. Instead of ignoring (and/or judging) these naïve and irresponsible practices, academic statisticians should come down from their ivory towers and invite industry to participate in networks of learning. Collaborative research projects will not only enrich the discipline, but also enhance the

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way statistics is applied, and advance proper assessment of methodologies in industry (also see Kettenring, 1997; Cobb & Moore, 1997).

Moreover, statistics has a value in enriching other sciences, such as physics, biology, economics and engineering. Interdisciplinary research can make statistics an integrating force among other sciences, which will be valued and celebrated. This will require statistics as a discipline to switch from a defensive, survivalist mode of thinking to planning new, innovative and courageous approaches to curriculum development.

Students' experiences

Many students enrol for statistics as a service module and never entertain the thought of pursuing the subject beyond first-year level. Ensuring that students' first contact with statistics at an introductory level is not their last, is the task of the academic statistician (see Moore, 2001). Kettenring (1997: 1230) proposes, as part of the "image reconstruction" needed by academic statistics, that the discipline can do more effective outreach and advocacy among students and the general public. New trends in statistical education should be investigated and validated in use. Kettenring (1997: 1231) proposes, for example, "holistic statistics' – moving away from "reductionist science" to a more "inclusive interpretation". To familiarise a student with analytical tools, probability and probability distributions, sampling and so on is no longer enough. Students must be challenged to develop the ability to solve common problems and explore the power of computer-intensive data analysis models (Ward & Fountain, 1996). Statistics is not just a skill – knowledge of responsible statistical analysis and interpretation of data is essential in the training of students, and for any developing country such as South Africa. The discourse in the teaching of statistics should not stall on issues such as the need for numerical literacy, proficiency in English and the lack of general access to new technologies.

Further research into the experiences of students taking statistics is crucial for enriching the discourses on the future of statistics. An example of such research is the project by Lee, Zeleke and Mavrotheris (2004) described in "A study of affective and metacognitive factors for learning statistics and implications for developing an active learning environment". North and Zewotir (2006) researched the "value" of teaching statistics to social science students, and Zewotir and North (2007) explored statistical education in the lives of prospective engineers in South Africa (see also Gordon, 2004).

Africanisation of the curriculum

The idea with 'Africanisation' is not to contest the generally accepted canon of statistics as a discipline. Nevertheless, the naïve use of international textbooks and resources in South Africa must be contested. As Petrina (2004: 81) points out, no curriculum or

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textbook is neutral or apolitical. Any curriculum is drafted in response to questions about what it means to be human in a particular context. This article does not petition for a cheap popularisation or Africanisation of the statistics curriculum, but for a *legitimate* response to questions asked in statistics in the African context. What knowledge, skills, qualities and values does statistics on the African continent require practitioners to have? Africanisation of the statistics curriculum will also imply taking race and gender seriously in the teaching of statistics, and also in ensuring retention of successful students. See, for example, the research done by Davidson, (2007) on the "burdens of race" in the teaching of statistics in an introductory statistics course in Australia.

Effective use of technologies

The world is on the brink of even more technological changes, in fields such as nanotechnology, robotics and cybernetics. Teaching statistics using these new technologies effectively and responsibly is a significant challenge to the South African academic statistician. While the pool of academic statisticians is shrinking, technology has transformed (and continues to transform) the world into a global, networked village. How does the academic statistician respond?

Distance learning, online journals, Web-based alternatives and virtual meetings between specialists can facilitate a process in which academic statisticians produce different statistical and theoretical responses to the burning questions of national and international import.

Adequate financial reward

Although adequate financial reward is an obvious solution, it is possibly also the most complex one. Why is being an academic statistician financially not adequately rewarding? The academic world can never compete with the private sector, yet governments should be aware of the consequences of the depreciation of disciplines. For many years, a similar situation existed in engineering departments, as registered engineers were not interested in the meagre salaries offered at academic institutions. One solution would be to allow the academic statistician to consult privately. The additional income could then compensate for deficiencies in the remuneration package. A further possible advantage of such an agreement is that the academic would be challenged to stay abreast of the latest developments, practices and applications of the subject in different fields.
Although remuneration is an important factor to consider, the distinctive characteristic of being an academic (which is often overlooked) is love of the discipline and teaching. Higher education should investigate creative ways to encourage, support and reward good teaching.

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Ongoing research

The external pressures on higher education for increased research outputs are escalating. University statisticians, who are academics in an academic environment, will have to research the problems raised in this paper, and improve both the quality of teaching and productivity, thus proving to various stakeholders that they are authentic scientists. Individual, collaborative and trans-disciplinary research will not only contribute to discipline-specific reflection, but will also increasingly celebrate the interdependency and relatedness of being human.

Conclusion

The present crisis in statistics as an academic career is complex and often paradoxical. There is an international and national need for applied statistical competencies. However, if there are not enough academic statisticians to facilitate the learning of these competencies, the need will not be met. The crisis is deepened by an increase in naïve and irresponsible applications of statistics. "Statisticians know that our greatest challenge is not to sell our graduates, but to sell our discipline" (Moore, 2000).

Statistics as a profession has to effectively expand the scope and reach of statistical science in problem-solving. Shewhart and Deming (online) state that "the long-range contribution of statistics depends not so much upon getting a lot of highly trained statisticians into industry as it does in creating a statistically minded generation of physicists, chemists, engineers, and others who will in any way have a hand in developing and directing the production processes of tomorrow".

Cloning academic statisticians is not a realistic (or appropriate) solution to the crisis. There are a number of challenges statistics as a discipline faces in addressing the need for more statisticians as *academics*. Statistics. and the role it should play in Africa and higher education in making a difference in people's lives, has a vibrant and challenging future – but only if we accelerate the recruitment of the next generation of innovative and responsive academic statisticians in service of humanity.

References

Barabba, V.C. 1990. What does the future hold for statisticians? *The American Statistician* 44 (2): 125–128.

Barnett, R. 2000. University knowledge in an age of super complexity. *Higher Education,* 40: 404–422.

Bickel, P. 2004. Comment on "A report on the future of statistics" by B.G. Lindsay, J. Kettenring and D.O. Siegmund. *Statistical Science* 19 (3): 387–413.

Cheung, P. 1998. Developments in official statistics and challenges for statistical education. *Statistics Singapore Newsletter* July: 1–8.

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Cloning fact sheet, human project information 2003. [Online] Available at url: http: // www.ornl.gov/TechResources/Human_Genome/elsi/cloning.html (accessed on 24 March 2008).

Cobb, G.W. and Moore, D.S. 1997. Mathematics, statistics, and teaching. *The American Mathematical Monthly* 104 (9): 801–823.

Davidson, M.A. 2007. Understanding the burdens of race at a predominantly white university: the experiences of underrepresented students in an introductory statistics course. Dissertation presented in partial fulfilment of the requirements for the degree Doctor of Philosophy in the Graduate School of the Ohio State University. [Online] Available at url: http: //www.stat.auckland.ac.nz/~iase/publications/dissertations/07.Davidson.Dissertation.pdf (retrieved 10 June 2008).

Friedman, J.H. 2001. The role of statistics in the data revolution. *International Statistical Review*, 69 (1): 5–10.

Gnanadesikan, R. 1990. Looking ahead: cross-disciplinary opportunities for statistics. *The American Statistician* 44 (2): 121–125.

Gordon, S. 2004. Understanding students' experiences of statistics in a service course. *Statistics Education Research Journal* 3 (1): 40–59.

Green, P.J. 2003. Diversities of gifts, but the same spirit. *The Statistician* 52 (4): 423–438.

Healy, M.J.R. 1978. Is statistics a science? *Journal of the Royal Statistical Society* Ser. A. 141: 385–393.

Kettenring, J.R. 1997. Shaping statistics for success in the 21st century. *Journal of the American Statistical Association* 92 (440): 1229–1234.

Lamel, J. 2003. Statistics for the citizens of Europe. *Statistics in Transition* 6 (3): 383–394.

Lee, C., Zeleke, A. and Mavrotheris, M.M. 2004. A study of affective and metacognitive factors for learning statistics and implications for developing an active learning environment. [Online] Available at url: http: //www.cst.cmich.edu/users/lee1c/carllee/papers/Study-of-Affective-factors-04.pdf (retrieved 11 June 2008).

Lehohla, P. 2005. Personal communication re SASA conference 31 October to 2 November, Grahamstown, South Africa: special session on a statistical training strategy for the SADC region.

Lehohla, P. 2008. SA plays key role sustaining Africa's statistical systems. *The Star*: 2. February 22.

Lindsay, B.G., Kettenring, J. and Siegmund, D.O. 2004. A report on the future of statistics. *Statistical Science,* 19 (3): 387–413.

Martin-Guzmán, P. and Cervera, J.L. 2002. The training of official statisticians: challenges and experiences. *ICOTS6,* pp. 1–6. [Online] Available at url: http: //www.stat.auckland.ac.nz/~iase/publications/1/4c3_mart.pdf (accessed 24 March 2008).

Mbeki, T. 2003. Stats SA must be strengthened. Census 2001. [Online] Available at url: http: //www.safrica.info/doing_business/businesstoday (accessed on 24 March 2008).

Moore, D.S. 1997. New pedagogy and new content: the case of statistics. *International Statistical Review* 65: 123–165.

Africa Education Review 5(2).indd 182 12/1/08 9:45:34 AM

Moore, D.S. 2000. Undergraduate programs and the future of academic statistics. [Online] Available at url: http: //www.amstat.org/meetings/jsm/2000/symposium.html (accessed on 24 March 2008).

Moore, D.S. and Olkin, I. 1984. Academic statistics: growth, change, and federal support. *The American Statistician* 38 (1): 1–7.

Moore, D.S. and Cobb, G.W. 2000. Statistics and mathematics: tension and cooperation. *The American Mathematical Monthly* 107: 615–630.

Moore, D.S. Undergraduate programs and the future of academic statistics. *The American Statistician* 55 (1): 1–6.

North, D. and Zewotir, T. 2006. Teaching statistics to social science students: making it valuable. *South African Journal of Higher Education* 20 (4): 503–514.

Ome, G. 2003. Send in the clones. [Online] Available at url: http: //www.law.indiana.edu/lib/pubs/news/0497.html (accessed on 24 March 2008).

Peiris, M.S. 2002. A way of teaching statistics: an approach to flexible learning. *CAL-laborate,* Vol 9. [Online] Available at url: http: //science.uniserve.edu.au/pubs/callab/vol9/peiris.html (accessed on 24 March 2008).

Petrina, S. 2004. The politics of curriculum and instructional design. *Interchange* 35 (1): 81–126.

Pike, D.J. 2002. The professional statistician: a policy for continuing professional development. *ICOTS6*. [Online] Available at url: http: //icots6.haifa.ac.il/PAPERS/5d4_PIKE.PDF (accessed on 24 March 2008).

Randles, R. 1996. SPAIG update. *ASQC Statistics Division Newsletter* 16 (5): 3.

Ridgway, J., Nicholson, J. and McCusker, S. 2007. Teaching statistics – despite its applications. *Teaching Statistics* 29 (2): 44–48.

Shewhart, W.A. and Deming, W.E. Statistical quotes. [Online] Available at url: http: //ourworld.compuserve.com/homepages/Rainer_Wuerlaender/statquot.htm (accessed on 24 March 2008).

Smith, T.M.F. and Staetsky, L. 2007. The teaching of statistics in UK universities. *Journal of the Royal Statistical Society* 170 (3): 581–622.

SPAIG. [Online] Available at url: http: //www.stat.ufl.edu/spaig.html (accessed on 24 March 2008).

Straf, M.L. 2003. Statistics: the next generation. *Journal of the American Statistical Association* 98 (461): 1–6.

Ward, J.H. and Fountain, R.L. 1996. More problem solving power: exploiting prediction models and statistical software in a one-semester course. *Journal of Statistics Education* 4 (3). [Online] Available at url: http: //www.amstat.org/publications/jse/v4n3/ward.html (accessed on 24 March 2008).

Zewotir, T. and North, D. 2007. Focus on the statistical education of prospective engineers in South Africa. *Pythagoras* 65: 18–23.

Africa Education Review 5(2).indd 183 12/1/08 9:45:34 AM